



AI Leadership in Action

Shared Understanding and Painpoints

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Content

What is AI, What is Not.

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Spectrum of Possibilities

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Demonstrations — The Showroom

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Simulations: Is this AI, does this work at TNM?

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TNM and Malawi Reality Check

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Discussions & Reflections on Day I

What is AI, What is Not.

Introduction — Artificial Intelligence



- “*Machines with Minds*”

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- **Tasks:** Mundane, Formal, Expert.

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- **Goals:** NLP, Robotics, Planning, Machine Learning, Knowledge Reasoning and/or Thinking, Computer Vision, Problem Solving, Game Playing, etc.

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- **Applications:** Education, Business, Economics, Society, Healthcare, Government, Warfare, etc.

Techniques (Statistical vs Rule-based)



- **Statistical**
 - ML Models (MLP/ANN, CNN, etc.) built on Good Data
 - Existing Models / APIs with New Applications

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- **Symbolic / Rule-based**
 - Formal Techniques
 - Deterministic vs Probabilistic Rules
 - Context-free / context-sensitive formal languages

Spectrum of Possibilities

Level 1 — Simple Technologies



- Data Collection and Automated Analytics (Google Forms, Survey Monkey, Trends)

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- Cloud and its Technologies : Online, you can draw, translate, transcribe, check sentiments of social media texts, etc.

Level 2 — Non-technical Topics and Issues around 4IR

- Ethics of 4IR Techs, Dangers of 4IR
- Social Impact of AI
- AI and Politics, Cyberwarfare
- AI and Governance
- Adoption, Conspiracies and Social Media
- Access to 4IR
- Data Bias
- Language bias / Under-resourced languages
- Terrorism and 4IR
- New Frameworks for Regulation



Level 3 — Research Methods, Software Tools, and New Data Sources

- New Data Sources (e.g. Google Datasets)
- Tools (e.g. Matlab, SPSS, RStudio, etc.)
- Research methods
 - Prediction Models : Logistics Regression, etc.
 - Data Mining and Visualisation
 - New Algorithms or Models
 - Digital Marketing, Social Media Analytics
 - Serious Games / Educational Games
- Big Data Sources
 - Social Media
 - Machines / Hardware



Level 4 — Off-the-shelves, APIs and Applications

- Chatbots (e.g. Chat-GPT)
- VR/AR (Dreams replaying, therapy VR, VR edu, VR entertainment, etc.)
- 3D Printing, Gesture-base Computing
- Blockchain and Cryptocurrency
- Apps for Deepfakes
- IoT
- Off-the-shelve Robots and Drones (e.g. Pepper, Photo-drones, etc.)
- Drones in Agriculture
- Voice AI Systems : Siri / Alexa
- AI Apps across Domains (Finance, Accounting, etc)



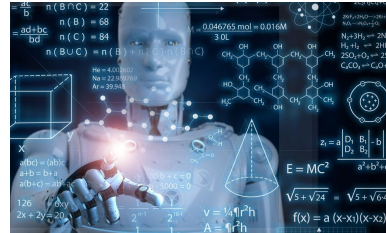
Level 5 — Data Science and Applications

- Big Data and Data Science
- Supervised Learning
- Nanotechnology
- Bio-Statistics
- Prediction : Earthquake, wealther, and other applications



Level 6 — Machine [Deep] Learning, New Algorithms and Tools

- Unsupervised and Deep Learning
- Quantum Computing
- Deep-fake (Detection)
- Surveillance : Analysis of Real-time Videos



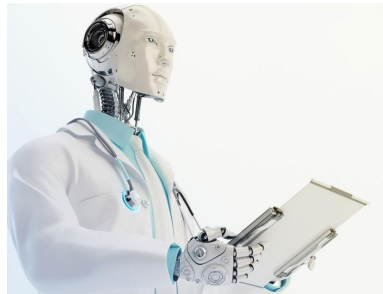
Level 7 — Expert Tools

- Game Playing AIs (e.g. Deep Blue)
- Robotic Surgery



Level 8 — Multidisciplinary Computation and Special Applications

- Cure/Treatment for Cancer
- Global Warming (Weather Control), etc



Level 9 — Advanced Drones and Machine

- War machines, military drones,
- Autonomous vehicles (DARPA, space navigation)
- Flying man
- Hyper-fast trains
- Space travel and exploration



Level 10 — The Dark Hole : Gene Editing and Super-soldiers

- Bio-weapons
- Genetic Modification of Babies
- Gene Editing, Edited/Super Soldiers
- Bio-Printing



Demonstrations — The Showroom

Simulations: Is this AI, does this work at TNM?

Game — What is AI, What is NOT!



Website

Game — Applied AI Demonstration Studio



TNM and Malawi Reality Check

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- **AI opportunities:** Predictive maintenance, capacity planning, energy optimisation and automated fault detection.

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- **AI opportunities:** Personalised offers, churn prediction, sentiment analysis and automated customer support.

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- **AI opportunities:** Fraud prediction, agent-liquidity forecasting, transaction-failure detection and personalised financial services.

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- **AI opportunities:** Forecasting, workforce analytics, document review, executive dashboards and decision-support systems.

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- **Network Experience Analytics:** Measuring and improving network performance.

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- **Fragmented Data and Slow Decision-Making:** Disconnected systems reduce insight.

Leadership Perspectives and Strategic Expectations

(Discussions, Form)

Pain-Point Diagnosis I & II

(Discussions: Organisational Reality, Prioritization, Impact)

Team Formation

(Any movements?)

Discussions & Reflections on Day I

Describe Day I — a word or phrase (Closed)

- Use one word, a phrase, or a short sentence to describe Day I.

Thank You — Please Let's Connect

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